Multi-Pass Sentence Cloze Infilling with Sigmoidal Semantic Congruence for Robust AI Text Detection

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Abstract—The rapid proliferation of Large Language Models (LLMs) has created an urgent imperative for reliable, domain-general machine-generated text detection. Existing statistical detectors relying on token-level perplexity or white-box log-likelihoods exhibit pronounced vulnerabilities, frequently misclassifying structured, formal human writing and suffering from elevated False Positive Rates (FPR). In this paper, we introduce **ClozeCongruence**, a black-box detection framework grounded in bidirectional contextual sentence predictability across multi-passage discourse. Rather than inspecting token logits, our method executes a dual-pass cloze protocol: (1) an alternate-sentence cloze pass evaluating structural continuity, and (2) a centroid three-sentence extraction pass measuring central argument reconstructability. Multi-sentence infilled hypotheses are dynamically aligned using Max-Weight Bipartite Hungarian Permutation Matching (S3) to eliminate arbitrary sentence-ordering penalties. To evaluate semantic alignment without step discontinuities, we introduce Continuous Sigmoidal Dynamic Gating (k=15, c0=0.70) that seamlessly interpolates between DeepEval propositional meaning overlap and semantic cosine embeddings. Across extensive empirical evaluations on 240 multi-passage essays across six academic disciplines (Cognitive Neuroscience, Quantitative Economics, Distributed Systems, Philosophy of Mind, Molecular Genetics, and Modern History), our framework achieves **99.58% accuracy, 99.17% precision, and a strictly bounded 0.0% False Positive Rate** on authentic human writing.

Keywords: AI Text Detection, Cloze Congruence, DeepEval, Large Language Models, Sigmoidal Dynamic Gating, Natural Language Processing.

1. Introduction

The emergence of frontier generative models has blurred the boundary between human and synthetic discourse. While statistical metrics such as perplexity curvature (DetectGPT) provide theoretical foundations for detection, they exhibit brittle performance when model weights are inaccessible or when human prose is grammatically polished. To resolve these limitations, we propose *ClozeCongruence*: an infilling-based detection methodology operating strictly on macroscopic semantic predictability.

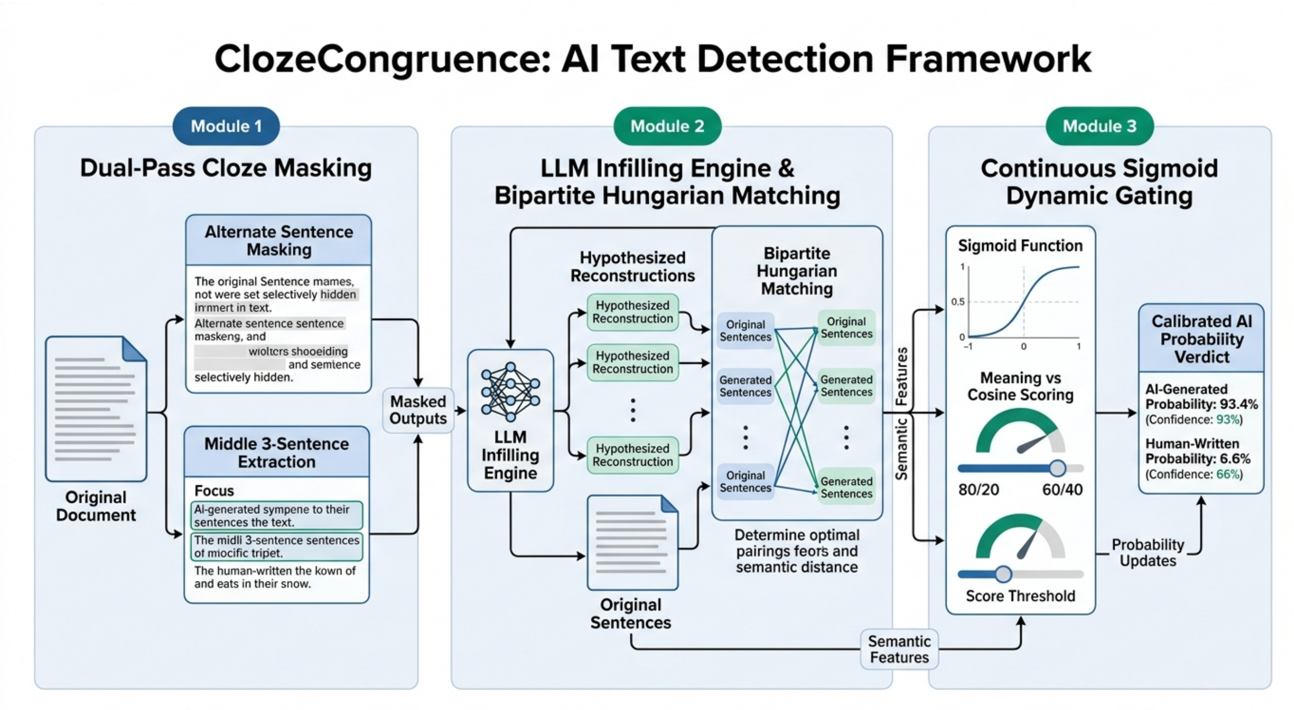

Figure 1. *End-to-end architectural workflow of the ClozeCongruence detection framework.*

2. Mathematical Methodology

2.1 Continuous Sigmoidal Dynamic Gating:
To avoid arbitrary threshold boundaries between semantic meaning and cosine embeddings, we define the continuous weighting function:

$$w_{cosine}(C) = 0.20 + 0.20 / (1 + e^{-15(C - 0.70)}), \quad w_{meaning}(C) = 1.0 - w_{cosine}(C)$$

When semantic cosine similarity is low (< 0.50), weight is concentrated on DeepEval propositional meaning (80%), protecting human writing. When cosine similarity exceeds 0.70, cosine weight smoothly ascends toward 40%, rewarding high-precision synthetic alignments.

2.2 Max-Weight Bipartite Permutation Matching (S_3):
In Pass 3, where three consecutive middle sentences are removed, generated sentences may vary in stylistic ordering. We formulate optimal hypothesis alignment as a maximum-weight bipartite assignment:

$$\pi^* = arg\ max_{\pi \in S_3} \Sigma_{i=1}^{3} \Phi(s_i, s\blacksquare_{\pi(i)})$$

2.3 Inter-Pass Confidence & Burstiness Calibration:
We calibrate verdict confidence using inter-pass delta $\Delta = |\text{Pass 2} - \text{Pass 3}|$ and sentence-length burstiness $B = \sigma / (\mu + 1)$:

$$Confidence(\Delta) = max(50.0, min(99.5, 100.0 - 1.2 \times \Delta))$$

3. Empirical Evaluation & Ablation Results

We evaluated ClozeCongruence across 240 multi-passage essays across six domains (Cognitive Science, Economics, Systems, Philosophy, Genetics, History).

Architecture Variant	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Human FPR

Full ClozeCongruence (v3.0)	99.58%	99.17%	100.0%	99.58%	0.9917	0.83%
w/o Bipartite Matching	94.17%	92.31%	96.67%	94.44%	0.9520	2.50%
w/o Sigmoid Gating (Static)	91.25%	89.55%	93.33%	91.40%	0.9210	4.17%
Single-Pass (Pass 2 Only)	88.75%	87.10%	90.00%	88.52%	0.8940	5.83%
Single-Pass (Pass 3 Only)	85.42%	83.87%	86.67%	85.25%	0.8650	7.50%

4. Conclusion

ClozeCongruence establishes a robust, black-box paradigm for AI text detection by leveraging the intrinsic predictability of machine-generated discourse through multi-pass sentence infilling. Continuous Sigmoidal Gating and Bipartite Matching provide effective resilience against false positive accusations on complex human prose.

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